SIMATS ENGINEERING

Short Answer Test-Application based Question

COURSE NAME: COMPUTER VISION
COURSECODE:ITA05

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)

Max Marks:100

Annexure A

Short Answer Test-Application based Question

Code of Conduct:

I Dharshan B (192312150) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Dharshan.

1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.

sampling converts a continuous image into discrete pixels; low sampling causes loss of detail and poor image quality.

2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.

Low light reduces the light reaching the sensor, producing dark and noisy images. Improve illumination and to enhance image quality.

3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

Aliasing occurs when the sampling rate is insufficient to represent fine image details. Increase the sampling rate to reduce artifacts

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

pixel resolution determines spatial detail, while intensity resolution determines the number of brightness levels.

5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.

Intermediate-level vision extracts features, while high level vision recognizes objects. These level help surveillance system detect and track objects.

6. Explain how image formation models help in improving image quality in applications such as autonomous driving.

Image formation models explain how light, surface, lenses and sensors create images.

7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Quantization converts sampled pixel values into discrete intensity levels. Higher quantization levels more brightness details in digital images.

8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Sensor electronics, heat and insufficient lightening can introduce image noise. use better sensor and filters to minimize noise.

9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Low-level vision acquires images, while intermediate-level vision extracts facial features.

10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

Low resolution hides object details and reduces visibility. Increase sampling and improve camera focus.

11. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications.

CCD sensor provides good image quality, while CMOS sensor consume less power and offer faster readout.

12. A digital image appears distorted due to improper image formation. Identify possible causes and suggest improvements based on image formation models.

Lens errors and poor focus cause image distortion. use camera calibration and proper alignment.

13. Explain how the features of Computer Vision systems enable applications such as automated surveillance.

Computer vision detects, recognizes and tracks objects. These features enable automated surveillance.

14. A system processes satellite images for land analysis. Explain how digital image fundamentals and acquisition methods influence the accuracy of results.

Resolution and sensor quality affect satellite image details. Proper acquistion improves land classification accuracy.

15. Given an application like OCR (Optical Character Recognition), explain how sampling, quantization, and image acquisition collectively affect performance.

Sampling affects character details, while quantization represents brightness. clear images improve OCR accuracy.

16. A high-resolution image consumes excessive storage and processing time. Explain how sampling and quantization can be adjusted to optimize performance without significant quality .

Downsampling reduces image size and processing time. maintain sufficient resolution to preserve details.

17. An image captured under uneven lighting shows poor contrast. Explain how the image formation process contributes to this issue and suggest corrective measures.

uneven illumination causes poor image contrast. use uniform lighting and contrast enhancement.

18. In a medical imaging system, explain how improper quantization can lead to loss of critical diagnostic information.

Low quantization loses important medical details. Higher bit depth preserves diagnostic information.

19. A self-driving car system relies on visual input. Explain how image acquisition and Computer Vision features contribute to reliable decision making.

computer vision detects lanes, vehicles and pedestrians. clear images support safer driving decisions.

20. An application requires detecting fine details in images. Explain how sampling rate and intensity resolution should be chosen to meet this requirement.

High sampling captures fine image details. Adequate intensity resolution preserves brightness differences.

21. An image captured at a low sampling rate shows jagged edges. Explain how sampling rate affects spatial resolution and suggest improvements.

Low sampling produces jagged edges. Increase sampling and apply anti-aliasing.

22. A digital camera produces images with banding effects. Analyze how quantization levels contribute to this issue and propose a solution.

Low quantization causes visible banding. Increase bit depth to improve intensity representation.

23. In a thermal imaging system, explain how image sensing affects temperature-based visualization accuracy.

Thermal sensors detect infrared radiation. calibration improves temperature measurement accuracy.

24. A blurred image is obtained due to motion during acquisition. Explain how image formation and sensing contribute to this effect.

Movement during exposure causes image blur. use a faster shutter speed.

25. A satellite image shows loss of fine terrain details. Explain how sampling and resolution affect feature visibility.

Low resolution hides small terrain features. Use higher-resolution satellite images.

26. In a biometric system, explain how image acquisition quality affects recognition accuracy.

Poor image quality reduces recognition accuracy. Use clear images and proper illumination.

27. An image appears pixelated when zoomed. Analyze the role of sampling and resolution in this effect.

Enlarging low-resolution images causes pixelation. Capture images at higher resolution.

28. A digital image shows reduced contrast due to poor sensing conditions. Explain how acquisition parameters influence this.

Poor lighting reduces image contrast. Improve illumination and adjust exposure.

29. In a machine vision system, explain how improper quantization affects defect detection.

Low quantization hides small defects. Increase bit depth to preserve details.

30. A camera captures images with varying brightness levels. Explain how image formation models influence intensity variations.

Lighting and surface reflectance affect brightness. Control illumination and exposure.

31. An image processing system fails to detect edges clearly. Explain how sampling and acquisition affect edge visibility.

Low sampling and noise reduce edge visibility. Increase resolution and reduce noise.

32. In an industrial inspection system, explain how image sensing impacts defect detection reliability.

Poor sensing reduces defect visibility. Use suitable sensors and proper lighting.

33. A low-bit image loses important details. Explain how quantization affects information preservation.

Low bit depth loses intensity details. Increase bit depth to preserve information.

34. A surveillance image appears noisy in low light. Analyze how sensing and acquisition conditions contribute to noise.

Low light makes image noise more noticeable. Improve lighting and apply noise reduction.

35. A digital image suffers from aliasing when resized. Explain how sampling principles apply in this scenario.

Improper resizing causes aliasing artifacts. Apply anti-aliasing before downsampling.

36. In medical imaging, explain how increasing sampling rate improves diagnostic accuracy.

Higher sampling capture finer details. Suitable resolution improves image analysis.

37. A system captures images with inconsistent quality. Explain how acquisition methods influence uniformity.

Changing lighting and camera settings causes inconsistent images. Standardize acquistion conditions.

38. A high-speed camera captures distorted images. Explain how image formation and sampling contribute to distortion.

Fast motion and improper exposure cause distortion. use Suitable exposure Settings.

39. In an OCR system, explain how poor sampling affects character recognition accuracy.

Poor Sampling reduces character clarity. use high-resolution images for accurate recognition.

40. A digital image shows quantization noise. Explain its origin and suggest mitigation techniques.

quantization noise occurs when intensity values are rounded. Increase bit depth to reduce information loss.

41. A vision system processes images with varying illumination. Explain how image formation models help normalize such variations.

uneven illumination causes brightness variations. use normalization and illumination correction.

42. An image appears overly smooth with loss of detail. Explain how quantization levels contribute to this issue.

low quantization removes subtle image details. Increase the number of intensity levels.

43. In a facial recognition system, explain how resolution impacts feature extraction accuracy.

Low resolution hides important facial features. Use high-resolution images for accurate recognition

44. A camera system produces images with color distortion. Explain how sensing and acquisition affect color representation.

Poor lighting and white balance distort colors. Calibrate the camera and adjust white balance.

45. A digital image has insufficient contrast. Explain how image acquisition and formation influence contrast levels.

Poor illumination reduces image contrast. Improve lighting and enhance contrast.

46. A system requires detecting small objects in images. Explain how sampling and resolution should be adjusted.

Low resolution hides small objects. Increase sampling rate and spatial resolution.

47. In satellite imaging, explain how quantization affects classification accuracy.

Low quantization loses land-cover details. Adequate bit depth improves classification accuracy.

48. A vision system fails in low-light conditions. Explain how image sensing impacts system performance.

Low light weakness sensor signals and increase noise. Improve illumination and use sensitive sensors.

49. A digital image contains staircase artifacts. Explain how sampling causes this effect.

Low sampling creates staircase-like edges. Increase resolution and apply anti-aliasing.

50. In an industrial system, explain how acquisition noise affects measurement accuracy.

Image noise causes industrial measurement errors. use calibrated sensors and noise reduction.

51. A camera captures images with motion blur. Explain how image formation and acquisition timing affect clarity.

Object movement during exposure reduces image clarity. use shorter exposure times.

52. A digital image has poor dynamic range. Explain how quantization influences intensity representation.

Limited dynamic range loses dark and bright details. use suitable sensor and adequate bit depth.

53. In a real-time vision system, explain how acquisition speed impacts processing efficiency.

Higher acquisition speed captures more frames per second. Balance frame rate with processing capacity

54. A system captures images with uneven resolution across regions. Explain how sampling contributes to this issue.

uneven sampling causes different levels of detail. use consistent sampling and camera calibration.

55. In medical imaging, explain how image acquisition affects diagnostic reliability.

Poor acquisition hides important diagnostic details. Use suitable sensors and proper exposure.

56. A digital image appears overly compressed. Explain how sampling and quantization contribute to information loss.

Downsampling and reduced quantization cause information loss. Use suitable compression settings to preserve details.

57. A vision system detects objects incorrectly due to poor acquisition. Explain the relationship between sensing and detection accuracy.

Blur, noise and poor lighting reduce detection accuracy. Improve focus and calibrate the camera.

58. A high-resolution image still lacks detail. Explain how image formation models influence clarity.

Blur and lens errors reduce image clarity despite high resolution. Improve focus and calibrate the camera.

59. In an automated inspection system, explain how sampling affects defect detection sensitivity.

Low sampling hides small defects. Increase resolution to improve defect detection.

60. A camera captures images with inconsistent intensity levels. Explain how quantization and sensing contribute to this variation.

Sensor variations and quantization cause intensity differences. Calibrate sensors and use adequate bit depth.